

Exercises

Problems 1–10 are Core Exercises. Problems 11–20 are Challenge Problems at approximately AMC 8 and MATHCOUNTS level. All twenty problems have been newly written for this book rather than copied from previous contests.

Part A. Core Exercises

1. Five numbers have average 24. Four of the numbers are 17, 21, 26, and 31. Find the fifth number.
2. A student has test scores 82, 87, and 91. What score is needed on a fourth test to make the average exactly 88?
3. A backpack originally costs \$60. Its price is reduced by 15%. Find the sale price.
4. After a 20% increase, a monthly fee is \$54. What was the fee before the increase?
5. A 10-liter solution is 30% juice. How many liters of pure water must be added to make the mixture 20% juice?
6. A container has 15 liters of a 40% salt solution. How many liters of pure salt solution must be added to make the mixture 50% salt?
7. Seven years from now, Leo will be three times as old as he was five years ago. Find Leo's present age.
8. The numbers of three kinds of books are in the ratio 2 : 3 : 7. There are 96 books altogether. How many books are in the largest group?
9. A runner travels 6 miles at a constant speed of 8 miles per hour. How many minutes does the trip take?
10. A car travels 150 miles. It travels the first 60 miles at 40 miles per hour and the remaining 90 miles at 60 miles per hour. How long does the entire trip take?

Part B. Challenge Problems

11. The average of eight numbers is 15. When the largest number is removed, the average of the remaining seven numbers is 12. What number was removed?
12. A store increases the price of an item by 25%. By what percent of the new price must the store reduce it to return exactly to the original price?
13. A tank contains 24 liters of a 25% solution. Some of the mixture is removed and replaced by pure concentrate. The final mixture is 40% concentrate. How many liters were replaced?
14. Nora is four years older than Eli. Six years ago, Nora was twice Eli's age. Find their present ages.

15. A box contains red, white, and blue tiles in the ratio 3 : 4 : 5. After 6 red tiles are added, the ratio of red tiles to the total number of tiles is 1 : 3. How many tiles were originally in the box?
16. A bus leaves a station at 9:00 A.M. traveling at 45 miles per hour. A car leaves the same station at 9:40 A.M. along the same road at 60 miles per hour. At what time does the car catch the bus?
17. A 120-point contest has two kinds of problems. Each regular problem is worth 4 points and each challenge problem is worth 7 points. There are 24 problems altogether. If all problems are answered correctly, how many challenge problems are there?
18. A restaurant bill consists of the meal price, an 8% tax on the meal, and then a 20% tip calculated on the meal plus tax. The final amount is \$64.80. What was the price of the meal before tax and tip?
19. A ball rebounds to $\frac{2}{3}$ of the height from which it falls. Its third rebound reaches a height of 16 inches. From what height, in inches, was the ball originally dropped?
20. A class has n students with average score A . One additional student with score s joins the class, and the new average is B .
 - (a) Derive a formula for s in terms of n , A , and B .
 - (b) Use your formula when $n = 19$, $A = 78$, and $B = 79$.
 - (c) Write one new question about averages that can be answered using the same idea.

Beyond the Chapter

These questions are for exploration rather than speed.

1. A quantity is increased by $p\%$. Derive a formula for the percent decrease needed to return to the original value. Test your formula with $p = 25$.
2. A container holds V liters of a mixture with concentration a . You add x liters of pure water and want concentration b , where $b < a$. Derive a formula for x .
3. Two travelers begin D miles apart and move toward each other at speeds u and v . Derive a formula for the meeting time. Explain why the relevant relative speed is $u + v$ rather than $|u - v|$.
4. Choose one problem from this chapter. Write a three-sentence summary: (1) what quantity you chose as the unknown, (2) what relationship produced the equation, and (3) what general lesson the problem teaches. Then change one condition and create a new problem of your own.

Answer Key

1. 25.
2. 92.
3. \$51.
4. \$45.
5. 5 liters.
6. 3 liters.
7. 11 years old.
8. 56 books.
9. 45 minutes.
10. 3 hours.
11. 36.
12. 20%.
13. 4.8 liters.
14. Eli is 10 and Nora is 14.
15. 48 tiles.
16. 11:40 A.M.
17. 8 challenge problems.
18. \$50.
19. 54 inches.
20. (a) $s = (n + 1)B - nA$. (b) $s = 98$. (c) Answers will vary.

Notes on the Exploration Problems

For the first exploration problem, an increase by $p\%$ multiplies the original amount by $1 + p/100$. If the required decrease is $q\%$, then

$$\left(1 + \frac{p}{100}\right) \left(1 - \frac{q}{100}\right) = 1.$$

Solving gives

$$q = \frac{100p}{100 + p}.$$

For $p = 25$, this gives $q = 20$.

For the second exploration problem, conservation of the ingredient gives

$$aV = b(V + x),$$

so

$$x = \frac{(a - b)V}{b}.$$

For the third exploration problem, the distance between the travelers shrinks at $u + v$ miles per hour, so

$$t = \frac{D}{u + v}.$$